

1. Administrative & Study Identification

Full Study Title: A Trial to Investigate Benralizumab in Children With Eosinophilic Diseases **Short Title/Acronym:** CLIPS **Protocol Number:** D3255C00004 **NCT Number:** NCT06512883 **Posting ID:** 478362929341363011 **Trial Status:** RECRUITING **Sponsor/Company Name:** AstraZeneca **Investigational Product:** Benralizumab (Fasenra) **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety, tolerability, and pharmacokinetics (PK) of repeat dosing of benralizumab in children with rare eosinophilic diseases. **Secondary Objectives:** To evaluate the pharmacodynamic (PD) properties, efficacy, immunogenicity, and long-term efficacy during the extension period.

2.2 Background & Rationale Rare eosinophilic diseases, such as Eosinophilic Granulomatosis with Polyangiitis (EGPA), are characterized by tissue-damaging inflammation driven by elevated eosinophil counts. The current standard of care heavily relies on long-term oral corticosteroids (OCS), which carry significant adverse side effects, especially in pediatric populations. This study is necessary to evaluate if benralizumab—an IL-5 receptor alpha-directed monoclonal antibody that depletes eosinophils—can safely induce disease remission and serve as a corticosteroid-sparing treatment in children ages 6 to <18.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional (Basket Study) **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration **Target N\$:** 14 patients (Estimated for the initial EGPA cohort; additional cohorts in other eosinophilic diseases may be added in future protocol amendments) **Number of Sites:** Multicenter, International (Europe, Canada, US, Rest of World) **Estimated Study Start:** 17-Apr-2025 **Estimated Primary Completion:** 20-Jul-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants aged 6 to <18 years at the time of signing the assent form and their caregiver signing the informed consent form. 2. Body weight $\geq$ 15 kg. 3. **EGPA Cohort:** The prescribed dose of oral corticosteroids (OCS) (>0.1 mg/kg/day, max dose of 50 mg/day) must be stable (no adjustment of the dose) for at least 4 weeks prior to baseline (Visit 2). 4. **EGPA Cohort:** If receiving immunosuppressive therapy, the dosage must be stable for at least 4 weeks prior to baseline (Visit 2). 5. **HES Cohort:** Documented HES diagnosis (persistent eosinophilia >1500 cells/ μ L on 2 examinations $\geq$ 1 month apart with organ involvement) and symptomatic active HES or recent flare.

4.2 Exclusion Criteria 1. Any current malignancy or history of malignancy. 2. History of anaphylaxis to any biologic therapy or vaccine. 3. Known, pre-existing, clinically significant endocrine, autoimmune, metabolic, neurological, renal, gastrointestinal, hepatic, haematological, respiratory, or any other system abnormalities. 4. Previous receipt of benralizumab in an interventional clinical study. 5. **EGPA Cohort:** Diagnosed with granulomatosis with polyangiitis (previously known as Wegener's granulomatosis) or microscopic polyangiitis. 6. **EGPA Cohort:** EGPA relapse (any deterioration in EGPA and/or organ-threatening EGPA rendering patients unstable) within 3 months prior to screening and through first administration of IP. 7. **EGPA Cohort:** Imminently life-threatening EGPA disease within 3 months prior to screening and through first administration of IP.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants $\geq$ 35 kg will receive benralizumab Dose-1; participants < 35 kg will receive benralizumab Dose-2. Administered once every 4 weeks (Q4W). **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 52-week open-label treatment period, followed by an open-label extension period of at least 1 additional year (52+ weeks), plus safety follow-up (SFU) weeks after last IP administration.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care oral corticosteroids (OCS) and stable baseline immunosuppressive therapies. **Prohibited:** Previous receipt of benralizumab in an interventional clinical study; live-attenuated vaccines during the biologic treatment window.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	1 to 4 weeks prior Informed

Consent/Assent, Medical History, Eosinophil Count validation | |
Baseline | Day 0 (Visit 2) | First SC Dose, PK/PD Blood Sampling,
Baseline Clinical Assessments | | **Interim** | Weeks 4 to 48 (Q4W) |
Safety Labs, PK/PD tracking, Efficacy Assessment, Q4W SC
injections | | **End of Study** | Week 52 | Final Year-1 Assessments,
Safety Follow-Up, Option to enter Open-Label Extension |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Number of Participants with Adverse Events (AEs)**: To evaluate the safety and tolerability of benralizumab. 2. **Serum Concentrations of Benralizumab**: To evaluate the pharmacokinetics (PK) of benralizumab.

7.2 Secondary Endpoints 1. **EGPA Cohort**: Percentage of Participants with Remission at Week 24. 2. **Immunogenicity**: Number of Participants with Positive Antidrug Antibody (ADA). 3. **Pharmacodynamics (PD)**: Change From Baseline in Peripheral Blood Eosinophil Count.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous clinical assessments during Q4W site visits and throughout the 52-week open-label and extension periods. **Serious Adverse Events (SAEs)**: Routine 24-hour expedited reporting to the sponsor and regulatory authorities. **Data Monitoring Committee (DMC)**: An independent Data Monitoring Committee (DMC) oversees the safety and scientific integrity of the trial, particularly given the pediatric population.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants receiving ≥ 1 dose of IP), Pharmacokinetic (PK) Set, and Efficacy Set.

Sample Size Justification: A target enrollment of $N=14$ children (initial cohort) to adequately characterize the PK/PD and safety parameters using weight-tiered subcutaneous dosing. **Handling of Missing Data**: PK data will be summarized using observed data without imputation; standard imputation rules (e.g., LOCF or MI) will be applied to efficacy endpoints where appropriate.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to verify adherence to protocol, especially regarding pediatric assent and

weight-based dosing. **Audit Trail:** Robust Electronic Data Capture (EDC) system utilizing eCRF locking and data clarification forms.

11. Study Identifiers & Governance

Primary ID: D3255C00004 **Registry Number:** NCT06512883 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** CLIPS **Official Regulatory Title:** Phase 3, Open-label Trial to Evaluate Safety, Pharmacokinetics, and Efficacy of Benralizumab in Children With Eosinophilic Diseases (CLIPS)

12. Clinical Framework (Eosinophilic Specific)

Primary Condition: Eosinophilic Granulomatosis With Polyangiitis (EGPA) **Diagnosis Threshold:** Confirmed vasculitis and granulomatosis for EGPA. **Medical Methodology:** Interleukin-5 receptor alpha (IL-5Rα) directed targeted biologic therapy. **Optimization Target:** Eosinophil depletion, disease remission, and OCS-sparing capability.

13. Study Procedures & Methodology

Management Pathway: Targeted biologic add-on to stable immunosuppressive/corticosteroid therapy. **Education Protocol (HCP):** Training on pediatric weight-based SC dosing limits and vigilant AE tracking. **Education Protocol (Patient):** Caregiver/Patient education on study compliance, injection site reaction monitoring, and daily diaries if applicable. **Quality Audit Mechanism:** Centralized continuous review of PK/PD relationships and clinical relapse triggers.

14. Observation & Follow-Up Schedule

Total Duration: At least 104 weeks (52-week initial treatment + \$ \ge 52\$-week extension period) plus Safety Follow-Up. **Onsite Visit Cadence:** Every 4 weeks (Q4W) for SC drug administration and safety labs. **Remote Monitoring Frequency:** Between scheduled Q4W clinic visits as needed. **Checklist Review Interval:** Concurrent with the Q4W injection schedule.

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]			____		[DD-MMM-YYYY]								
	Clinical Operations Lead			[Insert Name]			____		[DD-MMM-YYYY]								
	Lead Statistician			[Insert Name]			____		[DD-MMM-YYYY]								